Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering
4th Year 1st Semester Final Examination, 2022
Session: 2018-19
Course Code: EEE-459
Course Title: Compound Semiconductor and Hetero-junction Devices

Total Marks: 100 **Credit: 3.0** **Total Time: 3 Hours**

Part - A

*(Answer any **two** of the following questions)*

1. (a) Draw the following crystal structures.
   (i) Hexagonal close packed crystal structure
   (ii) Zinc blende crystal structure.
   (iii) Body centered cubic crystal structure 6
   (b) Sketch energy-band diagram of a metal-semiconductor junction with an interfacial layer and interface states. 5
   (c) Determine the theoretical barrier height, built-in potential barrier, and maximum electric field in a metal–semiconductor diode for zero applied bias. Consider a contact between gold (metal work function 5.1 V) and n-type germanium (electron affinity 4.13 V) doped to $N_d = 10^{16}$ cm^{-3} at T = 300 K. $N_C = 1.04 \times 10^{19}$ and $\varepsilon_r = 16$ 6
   (d) Explain the process of vapor phase epitaxy. 8

2. (a) Draw the band diagrams of a pN and a nN hetero-junction. 5
   (b) Derive the equation for the depletion region width of a hetero-junction. 7
   (c) For forward active mode derive the excess minority carrier concentration equation at base of a bipolar junction transistor 8
   (d) Determine the n-type doping concentration, and depletion region width in P-type side of semiconductor for an n-Ge and P-GaAs hetero-junction. Consider built in potential is 1.0 V and T = 300 K, n-type Ge electron affinity 4.13, N_{vn} = [unclear: 6 ×] 10^{18} cm^{-3}, $N_d = 10^{16}$ cm^{-3}, $n_i = 2.4 \times 10^{13}$ cm^{-3} and $\varepsilon_n = 16$. For P-type GaAs electron affinity 4.07, $N_{vP} = 7 \times 10^{18}$ cm^{-3}, $N_a = 10^{16}$ cm^{-3} and $\varepsilon_P = 13.1$ 5

3. (a) Define specific resistance. Explain which parameters affect specific resistance. 4
   (b) With proper band diagram explain multiple negative differential resistance in a Resonant Tunneling Diode. 6
   (c) Briefly explain the types of ohmic contacts? 3
   (d) Draw energy-band diagram of the following metal semiconductor junctions and mention whether it is Ohmic or Schottky then also find the ideal Schottky barrier height.
   (i) Al on top of n-GaAs
   (ii) Cu on top of n-GaAs
   (iii) Ag on top of p-GaAs
   (iv) Au on top of p-GaAs

   Here, Al, Cu, Ag, and Au have work functions of 4.06 eV, 4.98 eV, 4.74 eV, and [unclear: 5.47] eV respectively. GaAs has electron affinity of 4.07 eV and Bandgap of 1.519 eV. 12

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Part - B

*(Answer any **two** of the following questions)*

4. (a) Draw the typical structure of a Double Hetero-junction Bipolar Transistor Structure and their doping profile 6

   (b) Describe the advantages of HBT over BJT. Show that for an HBT, the ratio of J_{nE} and J_{pE} can be expressed as $\frac{J_{nE}}{J_{pE}} = \frac{N_{DE}}{N_{AB}} \cdot \frac{W_E}{W_B} \cdot \frac{D_{nE}}{D_{pE}} \exp\left(\frac{\Delta E_g}{kT}\right)$ [*sic*: should be D_{nB}]. 7

   (c) For an ideal Metal Oxide Semiconductor capacitor draw charge distribution, electric field distribution, potential distribution and capacitance versus voltage curve. 6

   (d) Show the relation between Flat band voltage and threshold voltage of a MOS device. 6

5. (a) Derive the equation for emitter injection factor of a bipolar junction transistor. 7

   (b) Briefly explain the following non-ideal effects of a bipolar junction transistor.

   (i) Early effect
   (ii) Current Crowding 6

   (c) Describe the Gummel-Poon model of a bipolar junction transistor. 7

   (d) Consider a uniformly doped silicon npn bipolar transistor at T = 300 K with the following parameters: $D_B = 23$ cm^2/s, $D_E = 8$ cm^2/s, $\tau_{B0} = 2 \times 10^{-7}$ s, $\tau_{E0} = 8 \times 10^{-8}$ s, $N_B = 2 \times 10^{16}$ cm^{-3}, and $x_E = 0.35$ µm. The recombination factor has been determined to be $\delta = 0.9975$. The required common-emitter current gain is $\beta = 150$. A minimum neutral base width of $x_B = 0.80$ µm can be fabricated. (a) Determine an appropriate neutral base width and the minimum emitter doping concentration, N_E, to meet this specification. (b) Using the results of part (a), what are the values of α_T and γ? 5

6. (a) What are the advantages of High Electron Mobility Transistor (HEMT)? 4

   (b) With proper band diagram explain how High Electron Mobility Transistor (HEMT) operates. 7

   (c) Sketch the equivalent circuit for Ebers-Moll model as used in BJT. 5

   (d) Show that maximum cut off frequency of a MESFET is related to doping concentration. 4

   (e) Determine the channel thickness of a GaAs MESFET to achieve threshold voltage $V_T = 0.1$ V. Consider an n-channel GaAs MESFET at T = 300 K with a gold Schottky barrier contact. Assume the barrier height is $\varphi_{Bn} = 0.89$ V. The n-channel doping is $N_d = 2 \times 10^{15}$ cm^{-3}. 5

Term Test 1 (its questions 1 and 2 are 7(a) and 7(b))

[Term Test 1, Course Code: EEE-459, Credit: 3.0, Session: 2018-19, Total Marks: 10, Total Time: 30 minutes]

7. (a) What is ohmic contact? Briefly explain the types of ohmic contacts. 4

   (b) Determine the theoretical barrier height, built-in potential barrier, and maximum electric field in a metal–semiconductor diode for zero applied bias. Consider a contact between gold (5.1 V) and n-type germanium (4.13 V) doped to $N_d = 10^{16}$ cm^{-3} at T = 300 K. $N_C = 1.04 \times 10^{19}$ and $\varepsilon_r = 16$. 6

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Term Test 2 (its questions 1 and 2 are 8(a) and 8(b))

[Term Test 2, Course Code: EEE-459, Credit: 3.0, Session: 2018-19, Total Marks: 10, Total Time: 30 minutes]

8. (a) Sketch the energy band diagram of the MOS capacitor under various bias conditions for both a p-type and s-type [*sic*: should be n-type] substrate. 5

   (b) A p-type Si substrate at T = 300 K was doped at $N_a = 10^{14}\ cm^{-3}$ It has $Q'_{ss} = 10^{10}\ cm^{-2}$ and $t_{ox} = 30\ nm$ Determine the threshold voltage of an MOS system with a gold gate. Consider at T = 300 K, p-type Si electron affinity 4.01, $N_v = 1.04 \times 10^{19}\ \text{cm}^{-3}$, $N_c = 2.8 \times 10^{19}\ \text{cm}^{-3}$, $n_i = 1.5 \times 10^{10}\ \text{cm}^{-3}$ and $\varepsilon_r = 11.7$ and gold p-Si work function $\varphi_{ms} = 0.07$ V. 5

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